

**Supplementary table 2 Observed list of genes associated with drug resistance mechanisms in MDR-TB isolates**

| <b>Genes</b>            | <b>Resistance mechanism</b>                                                                                                                                                          | <b>Gene function</b>                                              |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <i>aftB</i>             | Mutations in <i>aft</i> gene disrupt the arabinofuranosyl transferase activity, resulting in altered cell wall structure and reduced binding of ethambutol (E) [44].                 | Terminal beta-arabinofuranosyltransferase                         |
| <i>ahpC</i>             | Mutations in the <i>ahpC</i> gene, leads to overexpression of alkyl hydroperoxide reductase, which reduces KatG activity, associated with isoniazid (H) resistance [38, 40]          | Alkyl hydroperoxide reductase C                                   |
| <i>ald</i>              | Mutations in <i>ald</i> that cause loss of function, alter the alanine metabolism, resulting in cycloserine and terizidone resistance [45]                                           | Alanine dehydrogenase                                             |
| <i>alr</i>              | Mutations in <i>alr</i> that cause overexpression, increases the alanine racemase activity, resulting in cycloserine and terizidone resistance [46]                                  | Alanine racemase                                                  |
| <i>bacA</i>             | Mutations in <i>bacA</i> reduce aminoglycoside uptake, leading to increased resistance to streptomycin (S) and related drugs such as amikacin, kanamycin and capreomycin. [47]       | Vitamin B <sub>12</sub> transport ATP-binding protein <i>BacA</i> |
| <i>clpCI</i>            | Mutations in the <i>clpCI</i> gene alter the ATP dependant protease activity, resulting in pyrazinamide (Z) resistance [48]                                                          | ATP-dependent Clp protease ATP-binding subunit <i>ClpCI</i>       |
| <i>ccsA</i>             | Mutations in <i>ccsA</i> , encoding cytochrome c biogenesis protein, arise as compensatory changes following <i>rpsL</i> mutation and contribute to streptomycin (S) resistance [49] | Cytochrome c biogenesis protein <i>CcsA</i>                       |
| <i>eis</i>              | Mutations in <i>eis</i> , increase the expression of acetyltransferase, resulting in kanamycin resistance [50]                                                                       | Aminoglycoside acetyltransferase                                  |
| <i>dnaA</i>             | Mutations in <i>dnaA</i> reduce <i>katG</i> expression, decreasing activation of isoniazid (H) and leading to intermediate resistance [51]                                           | Chromosomal replication initiator protein <i>DnaA</i>             |
| <i>embA, embB, embC</i> | Mutations in <i>emb</i> genes, alter arabinosyltransferase activity, disrupting arabinogalactan synthesis in cell wall, resulting in ethambutol (E) resistance [38]                  | arabinosyltransferase                                             |
| <i>embR</i>             | Transcriptional regulator of <i>embCAB</i> operon. Mutations can result in ethambutol (E) resistance [52]                                                                            | Transcriptional regulator                                         |

|                         |                                                                                                                                                             |                                                      |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <i>ethA</i>             | Mutations in <i>ethA</i> reduce its monooxygenase activity , preventing activation of ethionamide (ETH) [53]                                                | FAD-containing monooxygenase                         |
| <i>fbiA, fbiB, fbiC</i> | Mutations in <i>fbi</i> genes disrupt F420 biosynthesis pathway, preventing prodrug activation and resulting in Delamanid resistance [54]                   | F420 biosynthesis                                    |
| <i>fgd1</i>             | Mutation in <i>fgd1</i> gene, prevent activation of drugs [55]                                                                                              | F420-dependent glucose-6-phosphate dehydrogenase 1   |
| <i>folC</i>             | Mutations in <i>folC</i> prevent formation of hydroxyl dihydrofolate by DHFS, resulting in resistance to para-aminosalicylic acid (PAS) [56]                | Dihydrofolate synthase / folylpolyglutamate synthase |
| <i>gyrA, gyrB</i>       | Mutations in the quinolone resistance regions of <i>gyrA</i> and <i>gyrB</i> alter dna gyrase, resulting in resistance to fluoroquinolones (FQ) [57]        | DNA gyrase subunit A, DNA gyrase subunit B           |
| <i>glpK</i>             | Mutations in the 7 cytosines homopolymeric tract of <i>glpK</i> disrupt glycerol kinase, causing slow growth and transient multi-drug tolerance in Mtb [58] | Glycerol kinase                                      |
| <i>inhA</i>             | Mutations in the <i>inhA</i> gene, reduce binding affinity of isoniazid-NAD adduct, resulting in resistance to isoniazid (H) [37]                           | NADH-dependent enoyl-acyl carrier protein            |
| <i>kasA</i>             | Mutations in <i>kasA</i> alter the $\beta$ -ketoacyl-ACp synthase enzyme, affecting mycolic acid synthesis, resulting in isoniazid (H) resistance [38]      | 3-oxoacyl-[acyl-carrier-protein] synthase 1          |
| <i>katG</i>             | Mutations in <i>katG</i> reduce or eliminate catalase-peroxidase activity, preventing activation of isoniazid (H) [39]                                      | Catalase-peroxidase                                  |
| <i>lpqB</i>             | Regulates the MtrAB signalling pathway supporting cell envelope maintenance and intrinsic resistance [47].                                                  | Lipoprotein <i>LpqB</i>                              |
| <i>Rv0678</i>           | Mutations in <i>Rv0678</i> cause upregulation of multisubstrate efflux pump <i>MmpL5</i> , resulting in resistance to clofazimine and bedaquiline [59]      | Transcriptional regulator of <i>MmpL5</i>            |
| <i>mshA</i>             | Mutations in <i>mshA</i> reduce mycothiol biosynthesis, affecting activation of ethionamide (ETH) [58]                                                      | D-inositol-3-phosphate glycosyltransferase           |
| <i>mtrA</i>             | Maintains cell envelope integrity which contributes to intrinsic resistance [47]                                                                            | DNA-binding response regulator <i>MtrA</i>           |
| <i>mtrB</i>             | Works with <i>mtrA</i> to regulate envelope integrity and promote intrinsic drug resistance [47]                                                            | Sensor histidine kinase <i>MtrB</i>                  |
| <i>nusG</i>             | Mutations in <i>nusG</i> reduce NusG-mediated                                                                                                               | Transcription termination /                          |

|                         |                                                                                                                                                                                        |                                                     |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
|                         | transcriptional pausing, resulting in rifampicin (R) resistance [43]                                                                                                                   | antitermination protein <i>NusG</i>                 |
| <i>panD</i>             | Mutation in <i>panD</i> gene reduces pyrazinamide (Z) binding and activity by altering its structure [61]                                                                              | Aspartate 1-decarboxylase                           |
| <i>pncA</i>             | Mutation in <i>pncA</i> gene reduces pyrazinamide (Z) activity by preventing its conversion to active form [62]                                                                        | Pyrazinamidase                                      |
| <i>rplC</i>             | Mutation in the <i>rplC</i> gene lead to modifications in the ribosomal protein L3 near the peptidyl transferase center, reducing linezolid binding [63]                               | 50S ribosomal protein L3                            |
| <i>rpoA, rpoB, rpoC</i> | Mutations in the genes induce conformational changes in Rna polymerase that reduce Rifampicin (R) binding [37]                                                                         | DNA-directed RNA polymerase subunits                |
| <i>rpsA</i>             | Mutations in <i>rpsA</i> gene alter the ribosomal protein S1, preventing the binding of pyrazinoic acid, resulting in pyrazinamide (Z) resistance [64]                                 | 30S ribosomal protein S1                            |
| <i>rpsL</i>             | Mutations in the <i>rpsL</i> gene alter the ribosomal protein S12, preventing the binding of streptomycin (S) [65]                                                                     | 30S ribosomal protein S12                           |
| <i>thyA</i>             | Mutations in the <i>thyA</i> gene alter the production of thymidylate synthase A, affecting the folate and thymine synthesis, resulting in resistance to para-aminosalicylic acid [66] | Thymidylate synthase ThyA                           |
| <i>thyX</i>             | Mutations in the <i>thyX</i> gene alter the folate pathway and contribute to para-aminosalicylic acid resistance [67]                                                                  | Flavin-dependent thymidylate synthase               |
| <i>tlyA</i>             | Mutations in the <i>tlyA</i> gene alter the activity of rRNA methyltransferase and prevent ribosomal methylation, resulting in resistance to capreomycin [67]                          | 16S/23S rRNA (cytidine-2'-O)-methyltransferase TlyA |
| <i>whiB6</i>            | Mutations in <i>whiB6</i> can alter function of transcriptional regulator and affect the ESX-1 dependant secretion, resulting in drug resistance [69]                                  | Transcriptional regulator                           |
| <i>whiB7</i>            | Mutations in the <i>whiB7</i> gene, lead to modifications in the pathways resulting in drug resistance [70]                                                                            | Transcriptional regulator                           |